determine, approximately, the diameter of a nucleus from the deflection of alpha particles in the substance in question, by ascertaining what fraction of them shows so strong a deflection that it cannot be accounted for by the effects of the external electric field of the nucleus, but can only be explained by direct collisions with the nucleus. The larger the nucleus, the more frequently will such collisions occur. These experiments have shown, for instance, that the diameter of a uranium nucleus consisting of 238 particles is about four times as large as that of the helium nucleus which consists of 4 particles. Their respective volumes are, therefore, as 1 to 4^3, in other words, the volume of the uranium nucleus is about sixty-four times the volume of the helium nucleus. However, the uranium nucleus has about sixty times as many component particles as the helium nucleus.

These two facts—the approximate equality of the binding energies of the individual particles, and the approximate proportionality of the number of particles to the volume of the nucleus—warrant the conclusion that the protons and neutrons are distributed approximately uniformly throughout the nucleus, because if this were not the case, their binding energies would show considerable differences, both in individual parts of a nucleus and in different nuclear types. Moreover, this fact, in conjunction with the proportionality of the number of particles and the volume, indicates that this density distribution is the same in all nuclei, except, again, for the very lightest atoms. Therefore, we may speak of a homogeneous nuclear substance in all nuclei, which consists of a mixture of protons and neutrons packed with an approximately constant density. There is a slight difference here between different atomic species, with respect to the ratio of the number of neutrons to the number of protons.

Therefore, we obtain a very accurate model of a nucleus by likening it to a drop of liquid. Just as water droplets of different sizes can form from water molecules, so drops of nuclear matter of different sizes—the different atomic nuclei—can form from protons and neutrons. This liquid-drop model has exactly the properties which can be observed in an atomic nucleus. For the molecules are packed with equal density of distribution throughout a drop of liquid, and all the molecules are bound together in it by the same quantity of energy. The knowledge of the existence of a universal, homogeneous nuclear substance facilitates greatly our understanding of nuclear structure.

But this liquid drop has also other, still more subtle characteristics, and now we must investigate whether these have their analogies in the nucleus. Actually, in a liquid drop, not all molecules are bound together equally firmly. The molecules on the surface are linked with the others on one side only, and therefore, on the whole they are less tightly bound than the others. This explains the phenomenon of *surface tension.* Certain considerations concerning energy, which are absolutely analogous to those which we have